

Module/framework/package	Name and brief description of algorithm	An example of a situation where using the provided GLM implementation provides superior performance compared to that of base R or its equivalent in Python (identify the equivalent in Python)
Base R	Iteratively Reweighted Least Squares (IWLS) serves as an optimization method which fits GLMs through successive updates of parameters by weighted least squares. The algorithm constructs working dependent variables to solve weighted least squares problems iteratively until it reaches convergence.	The algorithm suits traditional statistical analysis for medium-sized datasets when interpretability needs combination with statistical inference methods including p-values and confidence intervals. The package provides better functionality than statsmodels Python equivalents for detailed statistical diagnostics and model checking requirements.
Big Data version of R	Large datasets that exceed memory capacity can be processed through bigglm package through its incremental computation method. The packages sparklyr and SparkR enable R users to access Spark's GLM implementation through their interfaces. The R packages foreach and future and snow provide parallel capabilities that speed up multiple model fitting processes.	Users who need to process extensive datasets beyond memory capacity should consider this package because it provides R syntax compatibility with diagnostic tools. The biglm package delivers optimal results when statistical inference demands remain crucial for datasets that exceed memory capacity of one machine.
Dask ML	Generalized Linear Regression employs distributed optimization algorithms (most likely L-BFGS by default) to perform GLM parameter estimation across a cluster. Distributed matrix operations on Spark's DataFrame abstraction	The system delivers optimal performance for handling enormous distributed datasets spread across multiple clusters. SparkR delivers dramatically better performance than base R when working with datasets exceeding one terabyte which

	execute the optimization process.	allows complete data parallelism. SparkR integrates better with R programming tools than the Python-based version of Spark.
Spark R	L-BFGS and SGD use Limited-memory BFGS (L-BFGS) which constructs an approximation of the objective function as a quadratic without computing Hessian explicitly. The SGD algorithm performs parameter updates by processing small data portions simultaneously. The distributed versions of these algorithms exist in Spark.	The system shows its best performance during model training operations on massive datasets spread across multiple clusters. L-BFGS shows faster convergence than SGD but users should opt for L-BFGS when it becomes available. The implementation delivers superior performance than standalone machine algorithms for datasets larger than 100 GB while requiring parallel processing.
Spark optimization	Scikit-learn chooses among 'lbfgs', 'liblinear', 'newton-cg', 'newton-cholesky', 'sag', 'saga' as its multiple solvers based on the model type and regularization requirements. The default solver is LBFGS while specialized solvers are available for different regularizations.	This solver works well for datasets of medium size while also enabling integration with Python data science tools. When dealing with large datasets that require L1 regularization the 'saga' solver demonstrates optimal performance. SAG and SAGA solvers outperform base R for very large datasets on a single machine.
Scikit-Learn	The optimization framework of Dask ML supports distributed implementations of ADMM and L-BFGS and Newton's method and proximal gradient methods to work directly with Dask arrays/dataframes rather than memory-based structures.	Python users who handle datasets that exceed single-machine memory capacity or span across cluster environments will find Superior most useful. The system delivers excellent performance when dealing with data located within the Dask ecosystem while requiring parallel processing across cores or machines. The system delivers better

		performance than scikit-learn when working with datasets that exceed RAM capacity.
--	--	--